

Week 2, Day 1

Week 2 – Day 1: Define Lab Requirements

Task: Define the requirements and technical capabilities needed to build the cybersecurity home lab.

Summary:

Identify and document the **hardware, software, networking, security, and analysis requirements** for the home lab. Define how **Oracle VirtualBox** will host the virtual environment, **Kali Linux** will function as the security analyst/testing system, **Metasploitable2** will serve as the vulnerable target, and **pfSense** will provide firewall and network security controls. Define the **DMZ and internal network structure** and establish how **Wireshark** will be used to capture and analyze network traffic.

The task also establishes the **project scope, constraints, and essential MVP features** required for the lab to operate as a controlled cybersecurity testing and network analysis environment.

Expected Outcome:

A clear requirements definition that establishes **what the lab must contain, how the components will interact, what the lab will be used for, its limitations, and the minimum functionality required before implementation begins.**